

Ulcerative Infections:

Genital ulcer disease manifests as a breach in the skin or mucosa of the genitalia, usually caused by a sexually transmitted infection. Of these infections, herpes simplex virus type 2 (HSV-2) is the most common etiology in most geographic areas, followed by syphilis and chancroid.

Herpes Simplex Virus

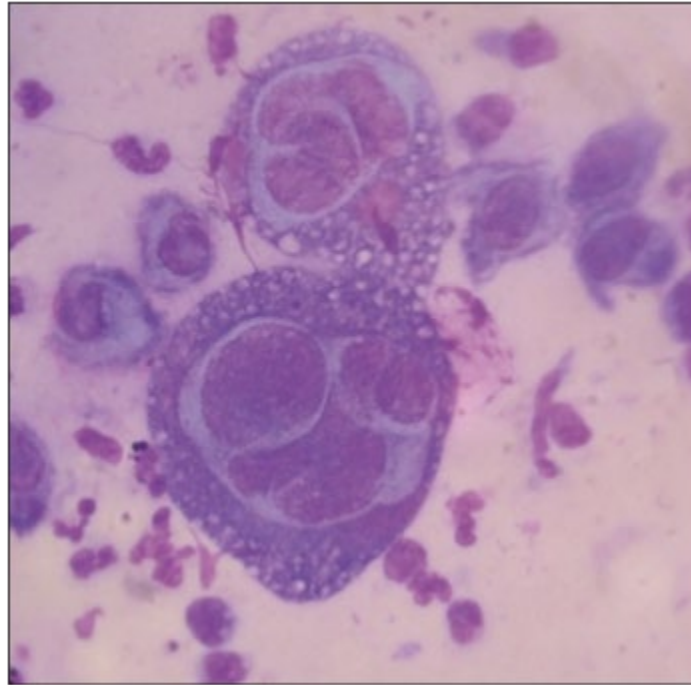
Genital herpes simplex virus (HSV) infection is common ulcerative lesion, and may involve the vulva, vagina, or cervix. HSVs are DNA viruses that include two serotypes, HSV-1 and HSV-2. HSV-1 typically results in perioral infection, whereas HSV-2 usually involves genital mucosa and skin; however, depending on sexual practices, HSV-1 may be detected in the genital region and HSV-2 may cause oral infections as well. By 40 years of age, approximately 30% of women are seropositive for antibodies against HSV-2.

Primary outbreaks are often asymptomatic, however about one-third of newly infected individuals are symptomatic. Lesions typically develop 3 to 7 days after transmission and are often associated with systemic symptoms such as fever, malaise, and tender inguinal lymph nodes. The earliest lesions usually consist of red papules that progress to vesicles and then to painful coalescent ulcers with an erythematous base. The lesions are easily visible on vulvar skin and mucosa, while cervical and vaginal lesions present with purulent discharge and pelvic pain. Lesions around the urethra may cause painful urination and urine retention. The vesicles and ulcers contain numerous viral particles, accounting for the high transmission rate during active infection. The mucosal and cutaneous lesions heal spontaneously in 1 to 3 weeks, but during the acute infection, the virus migrates to the regional lumbosacral nerve ganglia and establishes a latent infection. Because of viral latency, HSV infections persist indefinitely, and any decrease in immune function due to stress, trauma, concurrent viral infection, or hormonal changes can trigger reactivation of the virus and a *secondary outbreak* of the skin and mucosal lesions. As expected, recurrences are much more common in immunosuppressed individuals.



Transmission of HSV takes place mainly during the active phase but occasionally may occur during the latent phase due to subclinical virus shedding. Condoms and antiviral therapies reduce the risk of transmission, but do not prevent it completely. As with other sexually transmitted diseases, the virus is more readily transmitted to women than to men.

The diagnosis is based on typical clinical findings and HSV detection. Lesions are unroofed and the viral exudate collected. A Tzanck smear can be performed immediately, by spreading the material onto a slide and staining with Giesma. The viral cytopathic effect of multinucleated giant cells with intranuclear “ground-glass” appearance (Cowdry A inclusions) can be seen:



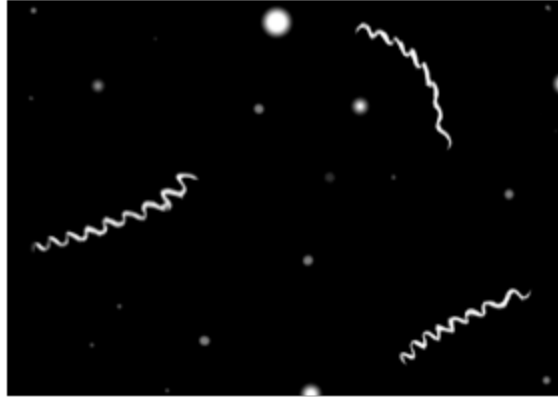
In addition, some laboratories offer more sensitive polymerase chain reaction tests, direct fluorescent antibody testing, nucleic acid amplification tests, and viral culture for detection. Individuals with primary, acute HSV infection do not have serum anti-HSV antibodies. Detection of anti-HSV antibodies in the serum is indicative of recurrent/latent infection.

There is no effective treatment for HSV; however, antiviral agents like acyclovir or famciclovir shorten the length of the initial and recurrent symptomatic phases.

Herpesvirus can be transmitted to neonates during passage through the birth canal of infected mothers. Although HSV-2 infection in the neonate may be mild, more commonly it is fulminating with generalized lymphadenopathy, splenomegaly, and necrotic foci throughout the lungs, liver, adrenals, and CNS. This risk is highest if the infection is active during delivery and particularly if it is a primary (initial) infection in the mother. Cesarean delivery is warranted in such cases to prevent transmission to the neonate.

Syphilis

Syphilis is a chronic sexually transmitted disease with varied clinical and pathologic manifestations. The causative spirochete, *Treponema pallidum*, can be visualized on silver stains, dark field microscopy, and immunofluorescence techniques. It is too slender to be seen on Gram stain and cannot easily be grown in culture.



Syphilis is divided into 3 stages

1. Primary: Approximately 3 weeks after infection. Characterized by a single localized, painless, ulcer known as a chancre. It is indurated with a clean base and rolled edge. Because it is painless, it may go unnoticed.



2. Secondary: 1-3 months later. Characterized by a maculopapular rash on palms and soles of feet, and painless lymphadenopathy (lymph nodes may feel rubbery). Condylomata lata are smooth, painless, wart-like lesions that appear on genitals.



3. Latent: Can last years. No symptoms occur, but positive serologic tests indicate continuing infection.
4. Tertiary: characterized by gummas (chronic granulomas), aortitis (vasa vasorum destruction resulting in aneurysms and aortic regurgitation), neurosyphilis (tabes dorsalis, general paresis), and Argyll Robertson pupils (constricts with accommodation but is not reactive to light).

Diagnosis is generally made serologically. Nonspecific serologic tests involve the use of nontreponemal antigens. These extracts react with antibodies in the serum samples of patients with syphilis. VDRL (Venereal Disease Research Laboratory) and RPR (rapid plasma regain) tests detect the presence of the antibodies. These tests are positive in most cases of primary syphilis and are almost always positive in secondary syphilis. False positives can occur with other infections and various autoimmune diseases, therefore positive results must be confirmed with treponemal specific tests. Fluorescent treponemal antibody absorbed test (FTA-ABS) is the most common specific treponemal antibody test. These tests remain positive for life even after effective treatment.

Penicillin G is the effective treatment for syphilis. More than half of patients with secondary syphilis will experience the Jarisch-Herxheimer reaction with treatment. This is characterized by fever, chills, myalgias, and other flu-like symptoms due to the lysis of treponemes and release of endotoxin-like substances.

Transplacental transmission occurs readily, and active disease during pregnancy results in congenital syphilis. In the infected neonates, skin and bone lesions, such as Hutchinson's teeth, mulberry molars, saber shins, saddle nose, snuffles, frontal bossing, and CN VIII deafness are common. Fetal infection can also result in stillbirth. All pregnant women should be tested in the first trimester and third trimester for syphilis, and treated with Penicillin promptly.



Chancroid

Chancroid is an acute, ulcerative STI caused by a small gram-negative rod called *Haemophilus ducreyi*. It is most common in tropical and subtropical areas among lower socioeconomic groups. It is one of the most common causes of genital ulcers in Africa, Southeast Asia, and the Caribbean, however is uncommon in the United States.

Clinically, chanroid presents as multiple, very painful, nonindurated ulcers with a gray or yellow exudate on the base. It is characterized by smaller, tender unilateral inguinal lymphadenopathy, which may rupture.



Diagnosis is difficult because it requires a special culture media and PCR tests are not widely available. Treatment is with Azithromycin

Lymphogranuloma venereum

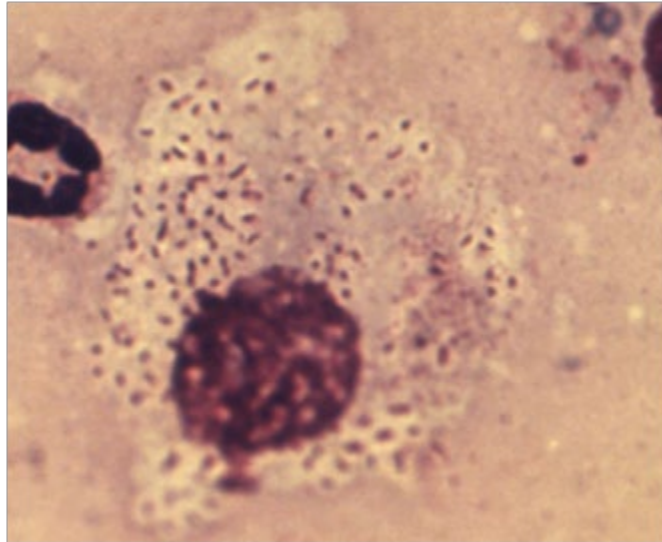
An infection of lymphatics, this disease is caused by *Chlamydiae trachomatis*. *C. trachomatis* has more than 15 immunotypes (A-L). Types D-K cause genital tract infections. Lymphogranuloma venereum is caused by immunotype L1-L3, and it presents as bilateral large and painful lymph nodes with fluctuant “buboes” and sinus tracts. It is accompanied by small, painless, shallow ulcers that are usually not observed.

It is diagnosed with NAAT and treated with doxycycline.



Granuloma inguinale

Also known as donovanosis, Granuloma Inguinale is caused by *Klebsiella granulomatis*. It is also uncommon in the United States, but is endemic in some rural tropical areas. It begins as a painless, beefy red ulcer with granulomatous appearance and rolled edges, that bleeds easily on contact. "Pseudobuboes" may occur in inguinal lymph nodes. Microscopy will show Donovan bodies on Giesma staining (dark staining oval organisms).



It is treated with doxycycline.

SUMMARY OF ULCERATIVE INFECTIONS

Syndrome	Pathogen	Appearance of Ulcer	Pain	Adenopathy	Diagnosis	Treatment
Genital herpes	HSV-2 (more common than HSV-1)	Multiple, small vesicles and ulcers with an erythematous base	Painful	Tender lymphadenopathy	Direct fluorescent antibody (DFA) testing and/or viral culture; sample taken from the base of the ulcer; Tzanck smear shows multinucleated giant cells; nucleic acid amplification test (NAAT) may be useful	Acyclovir, famciclovir, valacyclovir
Syphilis	<i>Treponema pallidum</i>	Single (usually) indurated ulcer with a clean base; self-resolves and may not be observed	Painless	Painless, regional lymphadenopathy; lymph nodes feel "rubbery"	Serologic screening with nontreponemal test (e.g., rapid plasma reagin [RPR] or Venereal Disease Research Laboratory [VDRL]); treponemal test (e.g., fluorescent treponemal antibody absorption [FTA-ABS]) for confirmation; dark field examination of fluid from the lesion if possible	Benzathine penicillin G
Chancroid	<i>Haemophilus ducreyi</i>	Multiple, nonindurated ulcers with a gray or yellow exudate at the base	Very painful	Tender, unilateral inguinal lymphadenopathy; lymph nodes may rupture	Difficult to diagnose; special culture media, if available, or polymerase chain reaction (PCR)	Azithromycin
Lymphogranuloma venereum	<i>Chlamydia trachomatis</i> serovars L1–3	Small, shallow ulcers that self-resolve and are not usually observed	Painless	Characteristic appearance of lymph nodes is key feature; may be bilateral, large, and painful; presents with fluctuant "buboes" and sinus tracts	NAAT; serology can also be used	Doxycycline
Granuloma inguinale (donovanosis)	<i>Klebsiella granulomatis</i>	Marked, beefy red, vascular ulcer with granulomatous appearance and rolled edges	Painless	Not a major feature; subcutaneous granulomas, "pseudobuboes" may occur	Difficult to diagnose; PCR if available; may also see dark-staining oval organisms on Giemsa staining (Donovan bodies) of biopsy specimen	Doxycycline

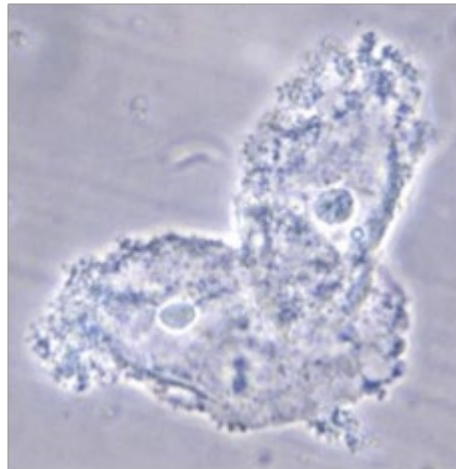
Vaginitis

Vaginitis is inflammation of the vagina that can result in discharge, itching, and pain. These symptoms occur primarily in three diseases: bacterial vaginosis (BV), candidiasis, and trichomoniasis.

Bacterial Vaginosis

BV is caused by an overgrowth of *Gardnerella vaginalis*. It is a gram-variable rod that presents as vaginal discharge. It is associated with sexual activity, but is not sexually transmitted. Diagnosis can be made by Amsel's criteria:

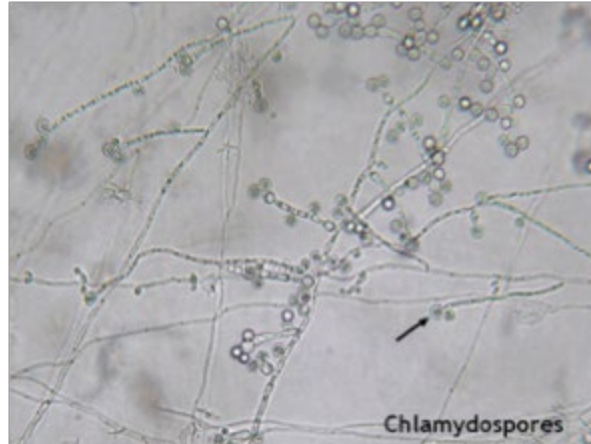
1. Characteristic discharge: thin, off-white, with fishy odor
2. Vaginal pH >4.5
3. Amine whiff test: mixing discharge with KOH enhances the fishy odor
4. Microscopy: showing clue cells, which are epithelial cells covered with Gardnerella causing a stippled appearance along the edge of the cell.



Treatment is with metronidazole. Of note, metronidazole can give a disulfiram-like reaction when taken with alcohol. It can present as nausea/vomiting, facial flushing, and abdominal pain.

Candidiasis

Vaginal candidiasis is caused by an overgrowth of *Candida albicans*. It traditionally presents as a clumpy, white, "cottage cheese" discharge that is odorless. It can be accompanied by vaginal itching and burning. It is associated with an acidic vaginal pH (<4.5). On microscopy, budding yeast and pseudohyphae will be visible.



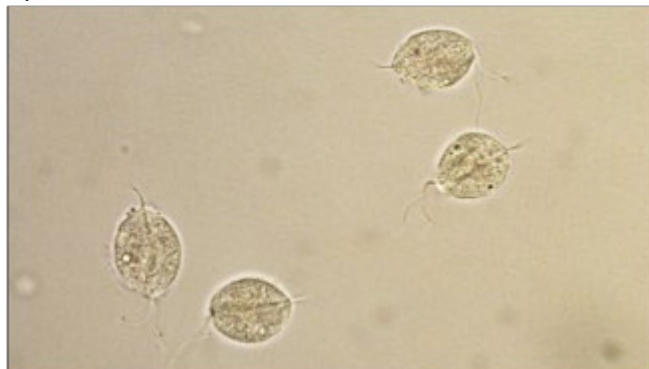
Treatment is with fluconazole (oral) or miconazole (vaginal suppository)

Trichomonas

Trichomonas is a sexually transmitted infection caused by the protozoa *Trichomonas vaginalis*. It often presents with a malodorous, frothy, yellow-green vaginal discharge, a vaginal pH >4.5, and punctate "strawberry" lesions on the cervix.



Microscopy will show motile protozoa.



Treatment is also treated with metronidazole.

SUMMARY OF VAGINITIS INFECTIONS

Bacterial vaginosis <i>(Gardnerella vaginalis)</i>	Trichomoniasis <i>(Trichomonas vaginalis)</i>	Candida vaginitis <i>(Candida albicans)</i>
 • Thin, off-white discharge with fishy odor • No inflammation	 • Thin, yellow-green, malodorous, frothy discharge • Vaginal inflammation	 • Thick, "cottage cheese" discharge • Vaginal inflammation
 • pH >4.5 • Clue cells • Positive whiff test (amine odor with KOH)	 • pH >4.5 • Motile trichomonads	 • Normal pH (3.8 - 4.5) • Pseudohyphae
Metronidazole or clindamycin	Metronidazole; treat sexual partner	Fluconazole

Cervicitis

Gonorrhea

Neisseria gonorrhoeae is a gram-negative intracellular diplococcus that results in a muculopurulent discharge with abundant neutrophils. Signs and symptoms develop within 3-5 days of infection, but asymptomatic infections are common in both men and women. Specimens can be tested by culture or NAAT. Untreated gonococcal infections can lead to pelvic inflammatory disease (discussed later), and disseminated gonococcal infection (DGI), with clinical signs and symptoms such as fever/chills, arthritis/arthralgias, tenosynovitis, skin lesions, endocarditis, and meningitis.

CDC just released updated treatment guidelines for 2021, which recommends ceftriaxone as monotherapy recommended treatment. It was previously recommended to treat with both ceftriaxone and azithromycin.

N. Gonorrhea can also be passed from mother to baby during birth, which can cause neonatal blindness. Erythromycin ointment is prophylactically placed in neonatal eyes as standard of care.

Chlamydia

Chlamydia trachomatis is a small gram-negative bacterium that is an obligate intracellular pathogen, because it cannot make its own ATP. As mentioned before, has more than 15 immunotypes (A-L). Types D-K cause genital tract infections. Genital infections by *C. trachomatis* are the most common bacterial STIs in the world. They are associated with similar clinical features as those caused by *N. gonorrhoeae* (muculopurulent discharge containing a predominance of neutrophils).

CDC just released updated treatment guidelines for 2021, which recommends doxycycline as monotherapy recommended treatment. Azithromycin can be used in pregnancy (this was previously the recommended treatment for all cases).

C. trachomatis can also be passed from mother to baby during birth, which can cause neonatal blindness. Erythromycin ointment is prophylactically placed in neonatal eyes as standard of care.

Pelvic Inflammatory Disease

PID is a polymicrobial infection of the upper genital tract (endometrium, fallopian tubes, ovaries, and pelvic peritoneum), as a result of direct spread of pathogenic organisms along mucosal surfaces after initial infection of the cervix. This can result in endometritis, salpingitis, and tubo-ovarian abscess.

The predominant organisms responsible for PID are *C. trachomatis* and *N. gonorrhoeae*. This can lead the way for other anaerobes to create a polymicrobial infection. Other organisms that have been isolated include *Mycoplasma*, *Streptococcus*, *Staphylococcus*, *Haemophilus*, *E. coli*, *bacteroides*, *peptostreptococcus*, *clostridium*, and *actinomyces*.

The greatest risk factor for PID is prior PID. Adolescence, having multiple sexual partners, not using condoms, and infection with any of the causative organisms are important risk factors.

Patients with PID may present with purulent cervical discharge, pelvic pain, fever/chills, elevated white blood cell count. Cervical motion tenderness is often found, characterized by the “chandelier sign” (where the patient “jumps to the ceiling” with palpation of the cervix).

Significant effects of PID include scarring of the tubes, which can result in a hydrosalpinx, tubal factor infertility, and risk of ectopic pregnancy. There can also be perihepatitis (Fitz-Hugh-Curtis syndrome), which results in inflammation leading to localized fibrosis and “violin string” adhesions of the anterior surface of the liver to the adjacent peritoneum. It is probably caused by chlamydial infection more often.

Mild symptoms can be treated with ceftriaxone or cefoxitin x1 dose followed by 14 days of oral doxycycline +/- metronidazole. If the patient fails outpatient treatment, is septic, or is unable to tolerate PO, then IV cefoxitin or cefotetan with doxycycline +/- metronidazole.

Sources:

Robbins and Cotran, Pathology

Medical Microbiology and Immunology